

LayoutSpec3D: 3D Scene Generation with Structured Scene Descriptions and Asset-Aware Relation Correction

Abstract—Text-driven 3D scene generation needs to turn object requirements, spatial relations, and static geometric constraints into usable layouts. Data-driven and large-model methods often yield omissions, unmet relations, collisions, and boundary violations after numerical layouts are grounded in 3D assets. We propose LayoutSpec3D, a configuration-driven framework for scenes with explicit object requirements, spatial relations, and boundary conditions and sufficient asset coverage. It converts text, scene dimensions, boundaries, and domain configurations into a structured scene description, then builds a 3D asset record from each selected asset’s geometry to rectify support, collision, and boundary constraints. The framework organizes outdoor scenes when requirements are specified and assets are available. LayoutSpec3D-FTCE (LayoutSpec3D Functional Topology Controlled Evaluation) fixes prompts and scene metadata, maps outputs to a common canonical scene, and applies shared final-scene checks. LayoutSpec3D delivers more complete and relation-consistent scene layouts while reducing geometric conflicts, including collisions, boundary violations, support errors, and floating objects. It further preserves intended spatial relations with minimal layout displacement and efficient generation.

Index Terms—3D scene generation, text-to-3D, structured scene description, asset-aware layout, geometric constraint rectification.

I. INTRODUCTION

3D scene generation constructs layouts from natural-language descriptions, scene dimensions, and boundaries for embodied intelligence, virtual environments, and digital content. Dense or function-oriented scenes need to meet object, spatial, and boundary requirements before rendering, interaction, or simulation. Across 3D environments, asset support, collision avoidance, and usable space determine whether an instantiated layout remains usable [1]–[4]. They also extend to outdoor scenes when the required assets are available.

Text-to-3D generation needs to instantiate language-derived counts, placement cues, and relations on specific objects. For example, wall attachment, tabletop placement, facing, adjacency, and passage clearance require object roles, support, orientation, distance, and boundaries. Their interpretation depends on the identities and roles of the participating instances rather than on categories alone. Prior work on functional and relation-based scene representations shows that object sets alone do not fully express functional organization [5]–[9]. Preserving the link from a textual requirement to its participating instances helps distinguish an omitted object from an unmet relation after layout generation. 3D-asset layout generation therefore needs an explicit correspondence among textual intent, object instances, and spatial relations.

Existing work mainly follows two paths. Data-driven methods learn joint distributions of objects and space from large scene collections [10]–[16], but their outputs are shaped by training data, object categories, and asset distributions. Language and vision–language models convert instructions into structured representations or numerical layouts [17]–[21]; these plans need consistency with final asset geometry. The handoff from a plan to selected assets is therefore more than a format conversion: it determines whether the intended instances and relations remain feasible in the final layout.

Structured scene representations, object identifiers, and executable descriptions make objects and relations checkable [22]–[25]. However, when requirements, initial layouts, and asset selection are maintained in separate stages, an omission, an unmet relation, and an asset-induced violation are difficult to attribute consistently. When the dimensions used to form an initial layout differ from those of the selected final assets, collisions, boundary violations, or relation drift arise; moving objects then disrupts functional relations [20], [26]–[28]. Planning, asset integration, and rectification need a shared instance-level representation.

Methods differ in prompts, randomness, asset sources, and output formats, which confounds raw comparisons [17], [20], [21]. For example, SceneEval jointly evaluates explicit object and relation requirements and spatial plausibility, including support and collision [29]. A common final-scene representation makes comparisons less dependent on native output formats and helps distinguish planning, asset integration, and geometric correction. 3D-asset layouts benefit from this representation, which separates object completeness, relation satisfaction, and static geometric validity.

These gaps motivate LayoutSpec3D, which targets scenes with explicit object requirements, spatial relations, boundary conditions, and sufficient asset coverage. Its scene description records object requirements and critical relations, and the 3D asset record stores the selected asset’s dimensions, coordinate conventions, and geometry. Static rectification checks support, collisions, boundaries, and relation drift. With the required configuration and assets, the framework organizes indoor and outdoor scenes and records relation preservation, rectification displacement, and runtime.

The main contributions are as follows:

- We propose LayoutSpec3D, an extensible framework for structured 3D scene generation. It applies to scenes with explicitly defined object requirements, spatial relations, and boundary conditions. After domain configuration and asset mapping, it organizes indoor and outdoor scenes with the corresponding assets.

- We establish a closed loop that combines a structured scene description with static rectification of instance-level 3D assets. The scene description explicitly records object requirements and critical spatial relations for subsequent checks. Using 3D asset records, the rectification module handles support alignment, collision resolution, boundary constraints, and relation recovery while limiting disruption to the original functional relations.
- We define LayoutSpec3D-FTCE (LayoutSpec3D Functional Topology Controlled Evaluation), a unified evaluation procedure for final scenes. Across different scene and functional configurations, it fixes prompts, random seeds, and scene metadata, converts outputs from each method into a common canonical scene, and reports object completeness, relation satisfaction, and static geometric validity under identical criteria applied after generation.

II. RELATED WORK

A. Data-driven 3D scene generation, primarily for indoor scenes

Data-driven 3D scene generation learns object co-occurrence, layout, and semantic priors from large scene collections. 3D-FRONT and 3D-FUTURE provide indoor layouts, furniture assets, and semantic annotations [10], [11]. GRAINS, Fast and Flexible Indoor Scene Synthesis, ATISS, and DiffuScene use recursive, convolutional, autoregressive, and diffusion models [12]–[15]. Later work adds relation priors, asset attributes, and explicit constraints [5]–[7], [30], [31]. These methods provide category and relation priors, but category-level examples do not identify the selected asset for a requested role or whether it fits the layout. LayoutSpec3D instead uses domain configurations, asset mapping, and an evaluable intermediate representation without learning a target-domain distribution.

B. Language-model layout planning and structured scene representations

Language and vision–language models translate textual intent into explicit plans. LayoutGPT uses CSS-like representations and in-context examples; DirectLayout separates top-view planning, 3D completion, and asset alignment; TreeSearchGen combines multi-level scene representations with global and local tree search [17], [20], [21]. Scene Language, Holodeck, AnyHome, InstructScene, LayoutVLM, SceneCraft, and Planner3D use scene languages, open vocabularies, semantic graphs, layout optimization, Blender scripts, or graph structures [6], [18], [19], [25], [27], [32], [33]. These representations make textual requirements executable, but category-level plans lack selected-asset dimensions, coordinate conventions, and support surfaces. Differences in final asset geometry invalidate planned placements or relations. LayoutSpec3D retains an instance-identified scene description through asset mapping and static rectification.

C. Asset-aware geometric validity and physical scene construction

3D assets differ from category-level dimensions, so numerical layouts need geometric checks. PhyScene checks

collisions, boundaries, and accessibility; DirectLayout and LayoutVLM use asset geometry for collision, boundary, and local-relation handling [20], [26], [27]. Research on instance segmentation, pose estimation, semantic metric mapping, and constrained registration likewise relies on instance identity and metric geometry [34]–[37]. Physics-constrained pose refinement models support relations during geometric adjustment, while collision detection motivates explicit conflict checks [38], [39]. Recent scene-aware asset retrieval work considers global coherence when selecting 3D assets for generated scenes [40], but it does not explicitly verify support contact, collision freedom, or boundary compliance after placement. SceneSmith and SceneWeaver include asset validation and scene construction [28], [41]. LayoutSpec3D focuses on post-instantiation static validity: the 3D asset record stores asset properties, and staged rectification addresses support, collision, boundaries, and relation preservation while recording preservation, displacement, and runtime.

D. Evaluation of text-to-3D scenes

Text-to-3D studies usually use their own scene collections, prompts, assets, and pipelines. DirectLayout compares indoor scene types, TreeSearchGen introduces 3DTindo-Bench, and SceneSmith studies house-scale realism, layout consistency, and runtime [20], [21], [28]. These differences make attribution to planning, asset mapping, or post-processing difficult. To mitigate this confounding, SceneEval evaluates explicit object and relation requirements together with spatial plausibility, including support and collision [29]. LayoutSpec3D-FTCE is a unified evaluation procedure for final scenes, not a benchmark: it fixes prompts, random seeds, and scene metadata, maps outputs to a canonical scene, and applies common geometric and relational checks.

III. METHOD

A. Problem definition

Given a text scene request, dimensions, boundaries, a domain configuration, and available assets, LayoutSpec3D treats text-to-3D generation as structured construction and instantiation. The domain configuration defines the object vocabulary, quantity requirements, functional zones, relation templates, and placement types. During language-model compilation, the request yields a structured scene description and an initial layout with shared instance identifiers. The description records declarative requirements, while the layout sets each instance’s position, scale, yaw, and anchor. The system records input, planning, asset-integration, and rectification states, tracing requirements, relations, and selected assets through the scene description and 3D asset record. This shared representation is consistent with prior scene representations that retain structured object relations and instance identifiers [22]–[24].

We formalize the structured scene description as

$$S = (D, V, Z, R, O, E, B, A, \Omega), \quad (1)$$

where D , V , Z , R , O , and E denote the domain, structural variant, functional zones, object categories and quantities,

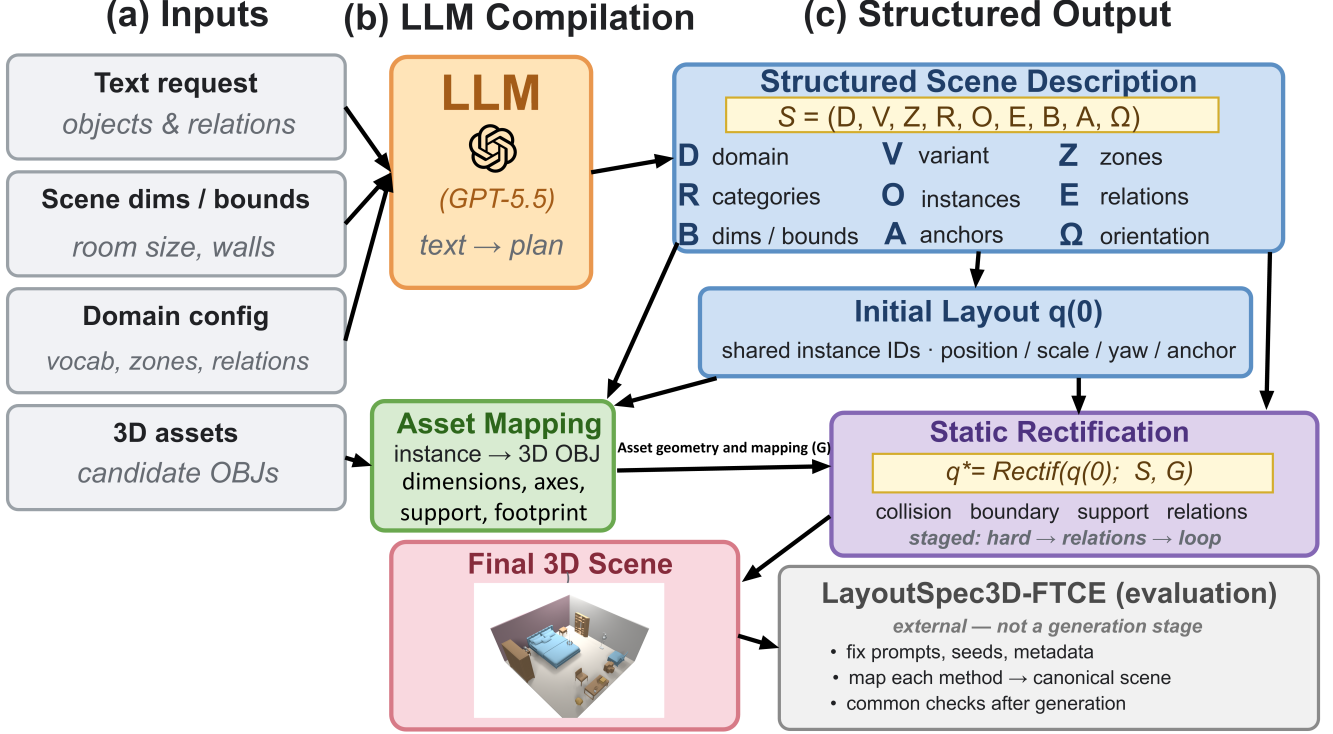


Fig. 1. Overview of LayoutSpec3D.

identified instances, and relations; B , A , and Ω specify dimensions and boundaries, placement anchors, and orientation conventions.

For scenes with explicit object requirements, spatial relations, boundaries, and sufficient asset coverage, let $\mathbf{q}^{(0)}$ be the initial layout produced with $\mathcal{S}$, and let G be the asset geometry and mapping information. Static rectification produces

$$\mathbf{q}^* = \text{Rectify}(\mathbf{q}^{(0)}; \mathcal{S}, G). \quad (2)$$

Here, $\mathbf{q}$ contains instance position, orientation, and support height; $\text{Rectify}(\cdot)$ is the relation-preserving static geometric rectification operator; $\mathcal{S}$ specifies types, quantities, zones, boundaries, anchors, orientation conventions, and relations; and G provides final-asset dimensions, coordinate axes, support types, and footprints. Rectification handles static support, collision, boundaries, and relation preservation; the interface accommodates later dynamics, accessibility, and fine-mesh checks.

B. LayoutSpec3D overview

Direct text-to-layout conversion risks conflating textual intent with later geometric errors. CSS-like layouts, scene languages, graphs, executable scene code, and other structured descriptions represent objects and relations [7], [17], [19], [22], [25], [33], [42], but do not isolate asset mismatches or geometric conflicts without retained instance identities and final geometry. LayoutSpec3D instead retains both in a structured description shared by asset mapping and rectification.

LayoutSpec3D-FTCE is a unified evaluation protocol, not a generation stage: it converts final outputs from all compared

methods to a canonical scene and applies common checks. LayoutSpec3D uses domain configurations, instance-level assets, and static constraints to construct the description, map assets, rectify geometry, and recheck relations. Its recorded stages separate planning, asset-geometry, and rectification changes.

C. Building a structured scene description

Planning representations do not replace final assets. Bounding boxes, CasaGPT’s cuboid primitives, and VoxScene’s discrete occupancy representation support planning [43], [44], but do not provide a target OBJ asset’s dimensions, coordinate conventions, support face, or rotated footprint. The scene description therefore assigns each object an identifier and records its category, target scale, placement anchor, and relation references. These fields drive asset mapping and localize later exceptions.

D. Instance-level asset mapping and geometry integration

After the scene description determines requirements and relations, LayoutSpec3D maps each abstract object to a 3D asset for rectification. Each instance is bound to an available asset from its category-specific candidate set. The selection stays fixed, even when same-category instances use different assets.

Each 3D asset record stores the semantic category, path, source dimensions, width–depth–height correspondence, up and forward conventions, target dimensions, scaling and origin

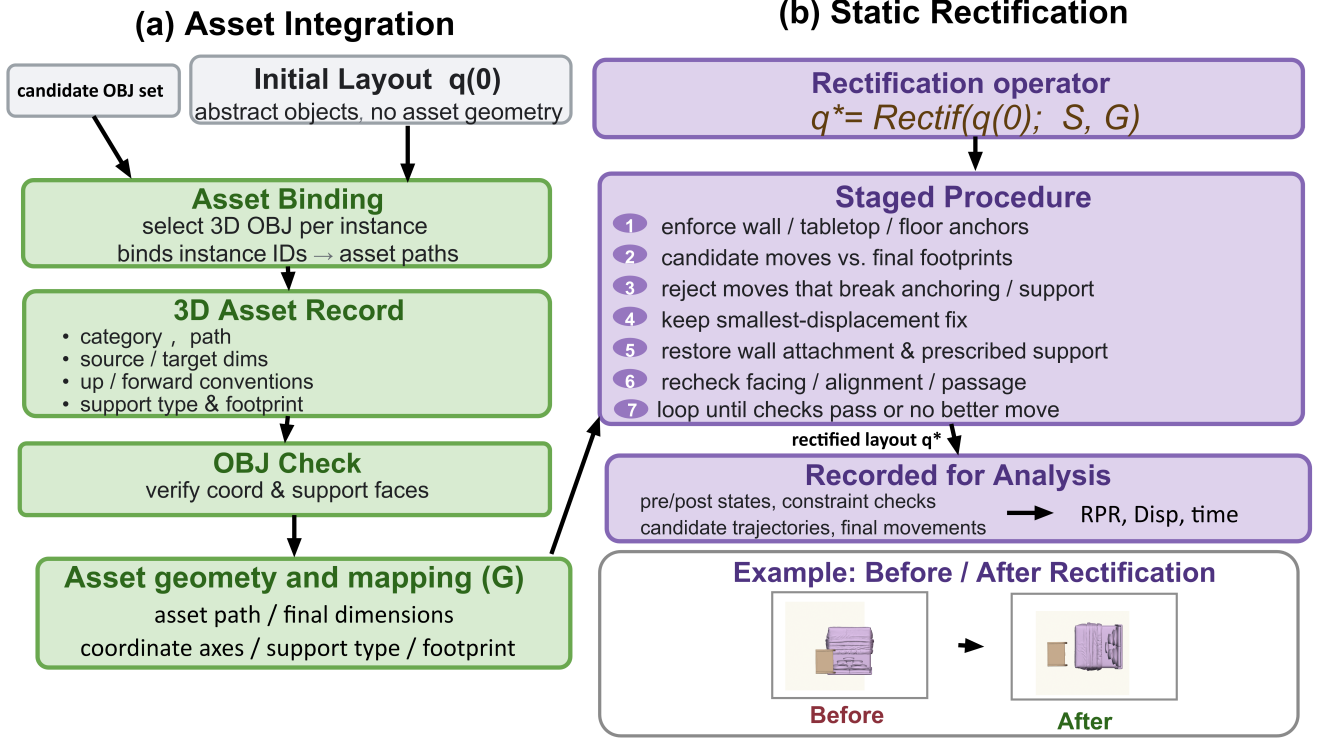


Fig. 2. Framework for instance-level asset integration and relation-preserving static geometric rectification. Initial-layout instances are bound to candidate 3D OBJ assets; the resulting records and OBJ checks supply asset geometry and mapping information for collision, boundary, and support checks, staged adjustment, and relation rechecking. Pre/post states, checks, trajectories, and movements are recorded for analysis.

policies, support type, and rotated footprint. The OBJ check verifies coordinate conventions and support faces. If the selected source lacks vertical or forward direction, a compatible fallback is used; otherwise, the binding is marked failed. The instance record then supplies retrieval, dimension, and collision information [28].

Final asset dimensions and yaw define footprints, collision bodies, and boundary checks. Rectification uses 3D-asset occupancy rather than planning boxes. The final record retains asset identity and geometry, distinguishing missing objects, mapping errors, and rectification failures.

E. Relation-preserving static geometric rectification

After asset instantiation, LayoutSpec3D applies staged, relation-preserving static geometric rectification. It removes geometric conflicts, retains functional relations, and limits unnecessary displacement:

$$q^* \approx \arg \min_{q \in \mathcal{F}(S, G)} [\lambda_c E_c + \lambda_b E_b + \lambda_s E_s + \lambda_r E_r + \lambda_m E_m]. \quad (3)$$

Here, $\mathcal{F}(S, G)$ contains states that satisfy the scene description and asset-geometry constraints. E_c , E_b , E_s , E_r , and E_m are the costs of collision, boundary violation, support violation, relation drift, and displacement; λ_c , λ_b , λ_s , λ_r , and $\lambda_m \geq 0$ set their relative priorities. The $\approx$ characterizes the staged objective rather than a global optimum. The system handles hard constraints before restoring relations.

Rectification is staged by relation type and asset anchor. It first enforces wall anchors, designated tabletop supports, and floor contact. Local candidate moves are checked against final footprints and constraints; moves that break anchoring or support are rejected, and the smallest-displacement move that reduces a collision or boundary violation is retained. Wall attachment and support are restored before facing, alignment, and passage relations are rechecked. The process ends when all checks pass or no candidate improves the current violation, retaining unresolved cases [43]–[45].

Pre- and post-modification states, constraint checks, candidate trajectories, and final movements are saved for ablations and relation-preservation, displacement, and runtime measures.

IV. EXPERIMENTS

We evaluate LayoutSpec3D through baseline comparisons, module ablations, and scope examples. The experiments examine object and relation satisfaction, geometric validity on jointly supported scenes, each core module, and configurations beyond the standard indoor comparison.

A. Experimental setup

To standardize language-model-dependent generation and reasoning, we use GPT-5.5 at each method’s language-model call sites. For LayoutSpec3D, GPT-5.5 only compiles the text request into an initial structured scene description and layout plan; asset mapping, geometry integration, static rectification,

canonical-scene conversion, and post-generation evaluation do not invoke a language model. The frozen LayoutSpec3D-FTCE configuration contains 105 prompts at three granularities and three fixed random seeds, giving each method up to 315 prompt-seed combinations. Comparisons share configurations, prompts, and seeds; outputs are converted to a canonical scene and checked after generation. Domains unsupported by a native implementation are excluded [17], [20], [21].

Quantitative comparisons use jointly supported indoor scenes, with shared dimensions, object requirements, relation descriptions, and asset interfaces. Common checks measure semantic requirements, geometric validity, relation preservation, displacement, and runtime. Values are percentages except Eff. Sem. Richness, Disp., and Time, following SceneEval and PhysScene [26], [29].

1) *Evaluation metrics*: Let $\mathcal{I}_m$ be the records for method m that produce an output and complete canonical-scene conversion. For record i , O_i is the final object set, and C_i and B_i contain objects in invalid collisions and out-of-bounds cases. An invalid collision is a non-support geometric intersection. We report semantic satisfaction, static geometric validity, and rectification preservation separately [26], [29].

Semantic satisfaction. For category c in the scene requirements, $n_i(c)$ and $n_i^{\text{req}}(c)$ denote the generated count and the required minimum count, respectively. Object recall and relation satisfaction are defined as

$$\text{O-Recall}_m = \frac{1}{|\mathcal{I}_m|} \sum_{i \in \mathcal{I}_m} \frac{\sum_c \min\{n_i(c), n_i^{\text{req}}(c)\}}{\sum_c n_i^{\text{req}}(c)}, \quad (4)$$

$$\text{Relation Sat.}_m = \frac{1}{|\mathcal{I}_{S,m}|} \sum_{i \in \mathcal{I}_{S,m}} \frac{\sum_{r \in R_i} \chi(r \text{ is satisfied})}{|R_i|}, \quad (5)$$

Here, $\chi(\cdot)$ is the indicator function, $\mathcal{I}_{S,m} \subseteq \mathcal{I}_m$ contains records with declared relations, and R_i is the relation set of record i . A relation is satisfied when all generated subject instances meet its wall-attachment, proximity, alignment, orientation, or support condition. Effective semantic richness counts unique categories among geometrically valid objects:

$$\text{ESR}_m = \frac{1}{|\mathcal{I}_m|} \sum_{i \in \mathcal{I}_m} |\{\text{cat}(o) \mid o \in O_i \setminus (C_i \cup B_i)\}|. \quad (6)$$

where $\text{cat}(o)$ denotes the semantic category of object o .

Static geometric validity. Col-obj and OOB-obj are mean within-scene proportions of invalid collisions and out-of-bounds objects; Col-scene and OOB-scene are record proportions with at least one case. Support-obj and Float-obj are prescribed-support violation and floating-object rates. Collision checks use final-asset footprints and vertical occupancy, and boundary checks use footprint and vertical extent. Valid support contacts are not collisions.

Support and floating status are computed from final canonical-scene oriented bounding boxes. For object o , let z_o and h_o be its center height and height, with bottom and top heights $z_o^- = z_o - h_o/2$ and $z_o^+ = z_o + h_o/2$; F_o is its rotated xy footprint. For supported object s and candidate carrier c , the contact gap and projected coverage ratio are

$$g(s, c) = |z_s^- - z_c^+|, \quad \rho(s, c) = \frac{\text{Area}(F_s \cap F_c)}{\text{Area}(F_s)}. \quad (7)$$

where $\text{Area}(\cdot)$ is horizontal-plane area; $\epsilon_z = 0.05$ m and $\tau_{\text{cover}} = 0.50$. Floor and valid inter-object contact are

$$\begin{aligned} \text{Floor}(s) &= \chi(|z_s^-| \leq \epsilon_z), \\ \text{Contact}(s, c) &= \chi(g(s, c) \leq \epsilon_z \wedge \rho(s, c) \geq \tau_{\text{cover}}). \end{aligned} \quad (8)$$

Support and floating. Each test case has floor or designated-object support targets. Let $T_i = T_i^{\text{floor}} \cup T_i^{\text{object}}$ be the targets in record i , s_t the target subject, c_t the declared carrier of an object-support target, and $q_i(t)$ its satisfaction indicator:

$$q_i(t) = \begin{cases} \text{Floor}(s_t), & t \in T_i^{\text{floor}}, s_t \text{ exists,} \\ \text{Contact}(s_t, c_t), & t \in T_i^{\text{object}}, s_t, c_t \text{ exist,} \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

The object-level support error rate is defined as

$$\text{Support-obj}_m = \frac{1}{|\mathcal{I}_{T,m}|} \sum_{i \in \mathcal{I}_{T,m}} \frac{1}{|T_i|} \sum_{t \in T_i} (1 - q_i(t)), \quad (10)$$

Here, $\mathcal{I}_{T,m} \subseteq \mathcal{I}_m$ contains records with nonempty T_i . A missing required object or carrier, floor separation, an excessive gap, or insufficient coverage is a support violation.

Float-obj tests physical support rather than designated-carrier use. Let O_i^{chk} contain objects in record i eligible for floor or object support checks, excluding floor decorations and objects with ceiling- or wall-attachment contracts. For $s \in O_i^{\text{chk}}$, define

$$p_i(s) = \chi(\text{Floor}(s) \vee \exists c \in O_i^{\text{chk}} \setminus \{s\} : \text{Contact}(s, c)), \quad (11)$$

The floating-object error rate is defined as

$$\text{Float-obj}_m = \frac{1}{|\mathcal{I}_{F,m}|} \sum_{i \in \mathcal{I}_{F,m}} \frac{1}{|O_i^{\text{chk}}|} \sum_{s \in O_i^{\text{chk}}} (1 - p_i(s)), \quad (12)$$

Here, $\mathcal{I}_{F,m} \subseteq \mathcal{I}_m$ contains records with nonempty O_i^{chk} . An incorrect carrier causes a Support-obj failure, but an object with physical support is not floating. Thus the metrics distinguish prescribed-support errors from actual lack of support.

Rectification preservation and process measures. Let R_i^{raw} be the index set of relations satisfied before rectification and R_i^{final} the corresponding set satisfied after rectification. The relation preservation rate is

$$\text{RPR}_m = \frac{1}{|\mathcal{I}_{R,m}|} \sum_{i \in \mathcal{I}_{R,m}} \frac{|R_i^{\text{raw}} \cap R_i^{\text{final}}|}{|R_i^{\text{raw}}|}, \quad (13)$$

Here, $\mathcal{I}_{R,m} \subseteq \mathcal{I}_m$ retains records with relations satisfied before rectification. For record i , M_i is the set of post-/pre-rectification object pairs from greedy category matching. In $(a, b) \in M_i$, a is the post-rectification object and b its matched pre-rectification object; p_a^{final} and p_b^{raw} are their horizontal center positions. Layout displacement is normalized by the room diagonal:

$$\text{Disp.}_m = \frac{1}{|\mathcal{I}_{D,m}|} \sum_{i \in \mathcal{I}_{D,m}} \frac{1}{|M_i|} \sum_{(a,b) \in M_i} \frac{\|p_a^{\text{final}} - p_b^{\text{raw}}\|_2}{\sqrt{L_i^2 + W_i^2}}. \quad (14)$$

Here, $\|\cdot\|_2$ is the two-dimensional Euclidean norm, $\mathcal{I}_{D,m}$ contains records with nonempty matching sets, and L_i and

TABLE I
QUANTITATIVE COMPARISON ON JOINTLY SUPPORTED STANDARD SCENES. VALUES ARE PERCENTAGES EXCEPT FOR EFF. SEM. RICHNESS, DISP., AND TIME.

Method	Eff. Sem. Richness $\uparrow$	O-Recall $\uparrow$	Relation Sat. $\uparrow$	Col-obj $\downarrow$	Col-scene $\downarrow$	OOB-obj $\downarrow$	OOB-scene $\downarrow$	Support-obj $\downarrow$	Float-obj $\downarrow$	RPR $\uparrow$	Disp. $\downarrow$	Time (s) $\downarrow$
DirectLayout	2.14	68.94	45.15	27.76	49.51	36.11	54.37	2.45	2.39	—	—	216.38
LayoutGPT	1.53	88.25	15.08	35.61	62.43	50.94	77.25	1.14	3.24	88.09	0.07	60.66
TreeSearchGen	0.64	44.09	59.32	17.01	32.28	37.54	36.61	1.95	2.84	—	—	569.66
LayoutSpec3D	2.34	95.76	66.21	9.09	23.39	15.92	14.21	0.92	1.17	96.48	0.02	48.57

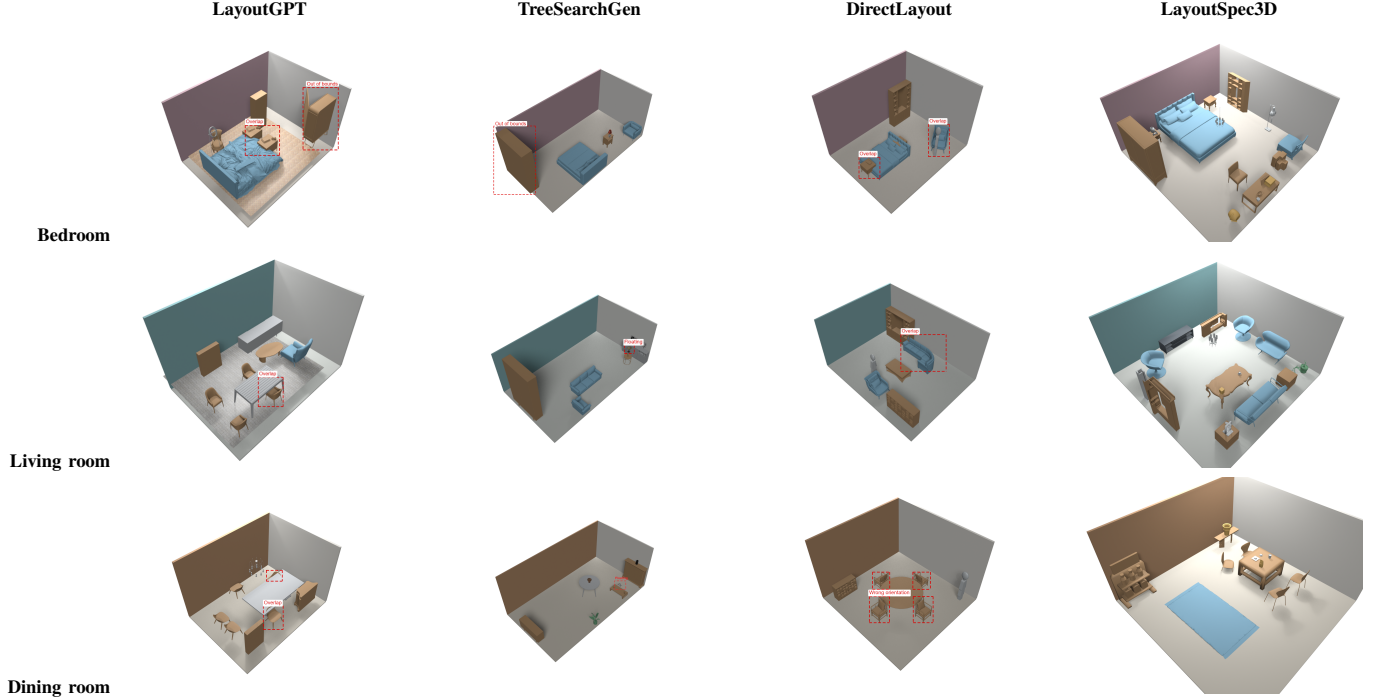


Fig. 3. Qualitative comparison of bedroom, living-room, and dining-room scenes. Each row is generated under identical object requirements, spatial relations, and boundary conditions.

W_i are scene length and width. RPR and Disp. require paired layouts before and after rectification. Time (s) is mean end-to-end runtime per scene over $\mathcal{I}_m$.

Ablations use matched standard configurations with unchanged prompts, assets, random seeds, and checks on the final scene. They remove structured scene description construction, static rectification, or both on the same test subset.

B. Comparison with baselines

Table I summarizes jointly supported scenes. LayoutSpec3D leads in effective semantic richness (2.34), object recall (95.76%), and relation satisfaction (66.21%), and has the lowest collision, out-of-bounds, prescribed-support, and floating rates. The scene description makes object and relation constraints explicit, while rectification checks geometric validity from 3D-asset footprints.

LayoutSpec3D attains 96.48% RPR with 0.02 displacement, compared with 88.09% and 0.07 for LayoutGPT. Its 48.57 s runtime is below LayoutGPT (60.66 s), DirectLayout (216.38 s), and TreeSearchGen (569.66 s). These are shared-environment measurements, not absolute software or hardware comparisons.

LayoutGPT reaches 88.25% object recall but 15.08% relation satisfaction, with 62.43% scene collisions and 77.25%

scene out-of-bounds cases. Object coverage alone therefore does not guarantee agreement with 3D-asset occupancy. DirectLayout is also lower than LayoutSpec3D on semantic, object, relation, and geometric measures.

TreeSearchGen reaches 59.32% relation satisfaction, but its object recall is 44.09% and runtime 569.66 s. Under shared configurations, LayoutSpec3D has stronger object and relation results with lower geometric-error and runtime measures.

For a qualitative comparison, Fig. 3 compares the same scenes under identical requirements, spatial relations, and boundaries. The baseline examples include incomplete objects, offsets, or occupancy conflicts; LayoutSpec3D uses its scene description and instance-level footprints for rectification.

C. Ablation study

Table II isolates the two modules. Without structured scene description construction, object recall falls from 95.76% to 67.36%, relation satisfaction to 55.25%, and RPR to 48.04%; later rectification does not restore unstructured planning information. Without static rectification, object- and scene-collision rates rise to 28.08% and 50.14%, out-of-bounds rates to 32.31% and 39.05%, Support-obj to 3.02%, Float-obj to 4.36%, and RPR falls to 66.05%. Footprint, boundary, and

TABLE II

ABLATION RESULTS FOR THE CORE MODULES. SCENE DESC. DENOTES STRUCTURED SCENE DESCRIPTION CONSTRUCTION AND RECT. DENOTES RELATION-PRESERVING STATIC GEOMETRIC RECTIFICATION. ✓ INDICATES A RETAINED MODULE AND – A REMOVED MODULE. ALL VALUES ARE PERCENTAGES.

Variant	Scene desc.	Rect.	O-Recall ↑	Relation Sat. ↑	Col-obj ↓	Col-scene ↓	OOB-obj ↓	OOB-scene ↓	Support-obj ↓	Float-obj ↓	RPR ↑
Full LayoutSpec3D	✓	✓	95.76	66.21	9.09	23.39	15.92	14.21	0.92	1.17	96.48
w/o Scene desc.	–	✓	67.36	55.25	11.53	25.08	16.84	15.00	1.45	2.25	48.04
w/o Rect.	✓	–	86.88	49.46	28.08	50.14	32.31	39.05	3.02	4.36	66.05
w/o Both	–	–	62.35	46.96	34.72	58.29	39.98	47.67	3.19	5.11	43.33

support constraints benefit from explicit treatment after asset integration.

Removing both modules reduces object recall and relation satisfaction to 62.35% and 46.96%, with 3.19% Support-obj, 5.11% Float-obj, and 43.33% RPR. The single-module ablations separate the roles: without the scene description, later adjustment does not recover absent structured relations; without rectification, object recall remains 86.88%, but collision, out-of-bounds, support-violation, and floating-object rates increase. The scene description therefore converts textual requirements into checkable instance targets before rectification removes local conflicts.

Fig. 4 shows the rectification process. Object relations and asset instances remain traceable before and after correction, consistent with Table II.

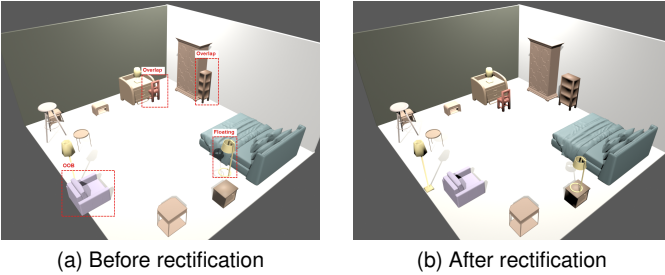


Fig. 4. Before-and-after comparison of static geometric rectification. Invalid overlap, out-of-bounds (OOB), and floating issues present before rectification are handled after rectification, while checkable object relations and asset-instance records are retained.

D. Qualitative results and scope

Beyond the standard indoor comparison, LayoutSpec3D also organizes scenes when object requirements, spatial relations, and boundaries are explicit and assets are available. Fig. 5 shows outdoor layouts with roads, buildings, and traffic objects. Comparable outdoor results are not included for DirectLayout, LayoutGPT, or TreeSearchGen under the shared protocol, so the examples illustrate scope rather than rankings against Table I.

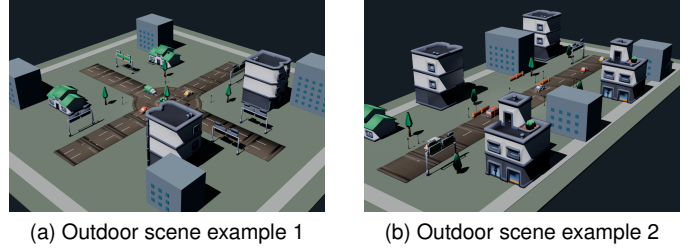


Fig. 5. Outdoor-scene examples within the scope of LayoutSpec3D. The scenes are jointly determined by available assets, object requirements, spatial relations, and boundary configurations.

The indoor tables compare generation and geometric quality where all methods are supported; the outdoor examples show reuse of the same interface under changed configurations. The scene description supplies categories, relations, and boundaries; the 3D asset record supplies dimensions, coordinate conventions, and footprints; static rectification checks local geometry.

V. CONCLUSION

LayoutSpec3D grounds textual requirements, spatial relations, and static geometry in 3D assets. The scene description records objects and relations; the 3D asset record links each selected asset to its geometry; rectification handles support, collision, boundaries, and relation drift. On jointly supported indoor scenes, it improves object requirements, relation satisfaction, and geometric validity. Ablations distinguish the modules: the scene description provides checkable objects and relations, while rectification resolves post-instantiation occupancy conflicts. With explicit requirements, relations, boundaries, and asset coverage, LayoutSpec3D organizes indoor and outdoor scenes. The interface is extensible to dynamic interaction validation, accessibility analysis, and fine-grained geometric checks.

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